Erol Akarsu

AI Software Engineer | Machine Learning Architect | Principal Software Engineer

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PROFESSIONAL SUMMARY

Hands-on engineer with 25+ years building enterprise search, backend services and cloud applications. Leads transformer-based product-image selection and generative project assistance at Lowe's. Combines production ownership with Python and TypeScript development, plus independent agent and RAG projects featuring grounded outputs, evaluation gates and human review. Ph.D. in Computer Science.

TECHNICAL SKILLS

Languages and applications: Python, TypeScript, Node.js, FastAPI, Flask, Java, Scala, React, Next.js, SQL

AI and retrieval: Transformer ranking, RAG, ChromaDB, grounded outputs, evaluation gates, model routing, context management, tool-call auditing, human-in-the-loop approvals

Data and search: PostgreSQL, Solr, Elasticsearch, Couchbase, Spark, Beam, Hadoop, Nutch, ETL

Cloud and delivery: Google Cloud, AWS, Docker, Kubernetes, Terraform, Jenkins, CI/CD, Pytest, Playwright

PROFESSIONAL EXPERIENCE

Lead Machine Learning Architect and Principal Software Engineer | Lowe's

Oct 2020-Present | ML architecture leadership since Jan 2024 | Charlotte and Mooresville, NC

- Lead architecture for a transformer-based system that selects product imagery for relevance and presentation quality.
- Build generative project-assistance capabilities that answer customer questions and support home-improvement planning.
- Led delivery of Lowe's.com free-delivery, expedited-delivery and promotional features backed by Spark processing.
- Improved search freshness with near-real-time Couchbase-to-Solr ingestion; built workflow applications using Beam pipelines and Elasticsearch services.
- Set architecture and release plans with product, engineering, design and senior leaders; balance customer impact, maintenance and compliance.

Senior Software Architect | Wipro Limited

May 2018-Oct 2020 | Richmond, VA

- Designed Lowe's enterprise search infrastructure on Google Cloud using Terraform, Packer, Ansible and Jenkins.
- Reduced indexing time for five million products from 360 minutes to 40 minutes by redesigning the Spark, Couchbase and Solr flow.
- Turned a 20-plus-step Spark and Scala indexing flow into a repeatable production pipeline with cloud infrastructure and deployment jobs.

Senior Software Developer | pExchange

Dec 2017-Mar 2018 | Richmond, VA

- Built role-based program-budget workflows using Spring Boot, MongoDB, Angular and Apache Shiro.

Master Software Engineer | Capital One

Aug 2016-Dec 2017 | Richmond, VA

- Led redesign of developer.capitalone.com and its regression suite using JavaScript, Node.js, AngularJS, Spring, Protractor and Cucumber.
- Automated API testing using Swagger and graph-based workflows; built reusable Spring and Spring Boot frameworks.

ADDITIONAL PROFESSIONAL EXPERIENCE

Independent Software Consultant | Self-employed

Sep 2003-Aug 2017 | United States

- Delivered government and commercial web, integration, data-processing and search projects from requirements through release.
- Built a Turkish retail search engine using distributed Nutch crawling, Hadoop, Solr, custom tokenization and 15 ranking criteria; search infrastructure covered 15,000+ sites and approximately 0.5 TB of indexed data.
- Reached 99.5% extraction success using XQuery rules on 700,000 lines of COBOL across 237 files; built secure service routing using password, certificate and token authentication.

Earlier Engineering Roles | Ferguson Enterprises and Cisco Systems

Nov 1999-Jul 2016 | United States

- Led web development and enterprise integration, with hands-on design, implementation and release support; worked on unified communications and supporting engineering systems.

SELECTED INDEPENDENT AI AND SOFTWARE PROJECTS

Working prototypes and reference applications, separate from the professional production systems above. Selected for agent control, retrieval, evaluation and operational reliability.

AI Agent Governance Control Tower. Built a Next.js, TypeScript and PostgreSQL control plane for agent registration, permission policies, tool-call auditing, human approvals, rollback and incident review. Persisted policy decisions and compliance evidence in durable records.

Company Brain. Implemented permission-aware answers over synchronized sources using connector ACLs, versioned documents and evidence snapshots. Added prompt-injection quarantine, typed grounded-answer contracts, evaluation gates and reviewer approval.

AI Model Cost Orchestrator. Designed routing around versioned model catalogs and evaluations, budget reservations, quality floors, circuit breakers and approval thresholds. Used React, Node.js, Flask, PostgreSQL and Kubernetes with tests and schema migrations.

Store Intelligence Platform. Created a RAG application using React, Flask, SQLite and ChromaDB vector search. Grounded store-operations recommendations in lead data and explicit business rules.

Personal Context MCP. Built tenant-scoped context-source workflows with explicit version transitions, permission review and optimistic concurrency. Protected mutations with idempotency keys and recorded changes in an append-only database audit trail.

MakePDF. Built a Python CLI and FastAPI service for verified redaction, bounded streaming intake, isolated OCR jobs, retention controls and signed downloads. Added immutable audit records and tests for document-handling failure paths.

SOLRProject Control Plane. Built a Python 3.12 discovery service that stages source changes in SQLite and applies replayable Solr jobs. Enforced ACL query filters and activated verified collection generations through atomic alias switches.

EDUCATION

Ph.D., Computer Science | Syracuse University | 1994-1999
M.S., Computer Science | Syracuse University | 1994-1996
M.S., Computer Engineering | Istanbul Technical University | 1991-1993
B.S., Computer Engineering | Ege University | 1986-1991

MACHINE LEARNING COURSEWORK AND CERTIFICATIONS

Neural Networks and Deep Learning; Machine Learning Regression and Classification; Advanced Learning Algorithms; Unsupervised Learning, Recommenders and Reinforcement Learning.